

NDN Community Meeting, September 10, 2020

Intel-NSF ICN-WEN Center Overview

Srikathyayani Srikanteswara



Intel/NSF Center on ICN-WEN

Information Centric Networking – Wireless Edge Networks

- \$6.5M over 3 years, 3 projects awarded
- Evaluate ICN's potential as a solution for 5G+ networks and applications for delivering ultra-low latency and high data bandwidths
- Evaluate realistic deployments and implementation complexities with applications including AR/VR and Autonomous driving
- Create a new, integrated ICN solution over wireless networks
- Develop solutions in three dimensions: (i) Wireless device endpoints, (ii) Network infrastructure, and (iii) Security and privacy

Overview of Projects

- **Title:** ICN-Enabled Secure Edge Networking with Augmented Reality
- **PI:** Lixia Zhang
- **Participating Universities:** UCLA, New Mexico State University, FIU
- **Title:** SPLICE: Secure Predictive Low-latency Information Centric Edge for next-generation wireless networks
- **PI:** P. R. Kumar
- **Participating Universities:** Texas A&M, Washington University St. Louis, Purdue University, Ohio State Univ., University of Illinois Urbana-Champaign
- **Title:** Light-Speed Networking (LSN): Refactoring the Wireless Network Stack to Dramatically reduce Information Response Time
- **PI:** Arun Venkataramani
- **Participating Universities:** Univ. of Mass (Amherst), U. of Wisconsin-Madison

ICN-WEN Advisory Team

- Board of Advisors

- (Intel) Geng Wu, Udayan Mukherjee, Eve Schooler
- (External) Dave Oran, Christian Tschudin

- NSF

- Thyaga Nandagopal, Darleen Fisher, Monisha Ghosh

- Intel

- Program management: Richard Chow
- Intel PI: Srikathyayani Srikanteswara
- Managing Sponsors: Jeff Foerster, Nageen Himayat
- Multiple Intel Champions

Learnings and Inspiration for Future Work

- In industry ICN gets compared to “existing systems” – not just TCP/IP
- Solve a real world problem for which there is no solution instead of offering a “better” solution
- Demonstrate how it can be integrated with existing systems and networks
- Enhance support for NDN over wireless, dynamic computing

Develop NDN based solutions for dynamic composable edge computing

